When the task fails, what may the agent change?

A decision theory of recovery, evidence, authority and stopping.

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RESEARCH QUESTION

Can an agent stay useful after a route fails, treating the failure as evidence about feasibility and never as a grant of new authority?

No new model runs. Library counts, archived runs and toy calculations are recomputed at build time.

Outline

01 **Incidents** — slides 3–8
Six reported cases in the style of the paper plan: framing, the feedback-loop latency, the Medicare chain, the substitution and scope-inference families, and the three questions a theory must answer.

02 **Decision theory** — slides 9–13
A constrained POMDP whose grant only a trusted reply can change, then four exact results: stopping with learning, value of information, ambiguity under a silent grant, selection from finite evidence.

03 **Algorithms** — slides 14–17
GRANT-LOOKAHEAD (online) and INCIDENT-SELECT (offline) as pseudocode with properties; a point-by-point comparison with POMCP, constrained RL, shielding, CaMeL, Progent, AI Control and ImpossibleBench.

04 **Experiments** — slides 18–20
E1 simulator regret study; E2 repair × grant factorial with seven arms including prior-method baselines; E3 selection contrast on fresh lineages; falsifiers for each.

05 **Implications** — slides 21–22
Each incident family mapped to a result, a system change and its test; outcomes interpreted in advance; milestones.

A failed route is where authority gets reinterpreted.

26 public incident records, October 2025 to September 2026, coded by pathway, mechanism, leading indicator and stop behaviour.

26 public incidents

Largest pathway: live-access overreach, 9 of 26. Repeated task failure recorded in 9; unknown in 13.

One decision point

Task blocked. Allowed route failing. Forbidden resource reachable. What changes: the plan, or the permission?

160 archived runs

Local service fixture, effect oracle in SQLite. 13 scorer passes carried an unauthorized effect; 2 violations never passed.

A theory and two algorithms

A constrained POMDP whose grant only a trusted reply can change; an online planner and an offline selector; tests that can fail.

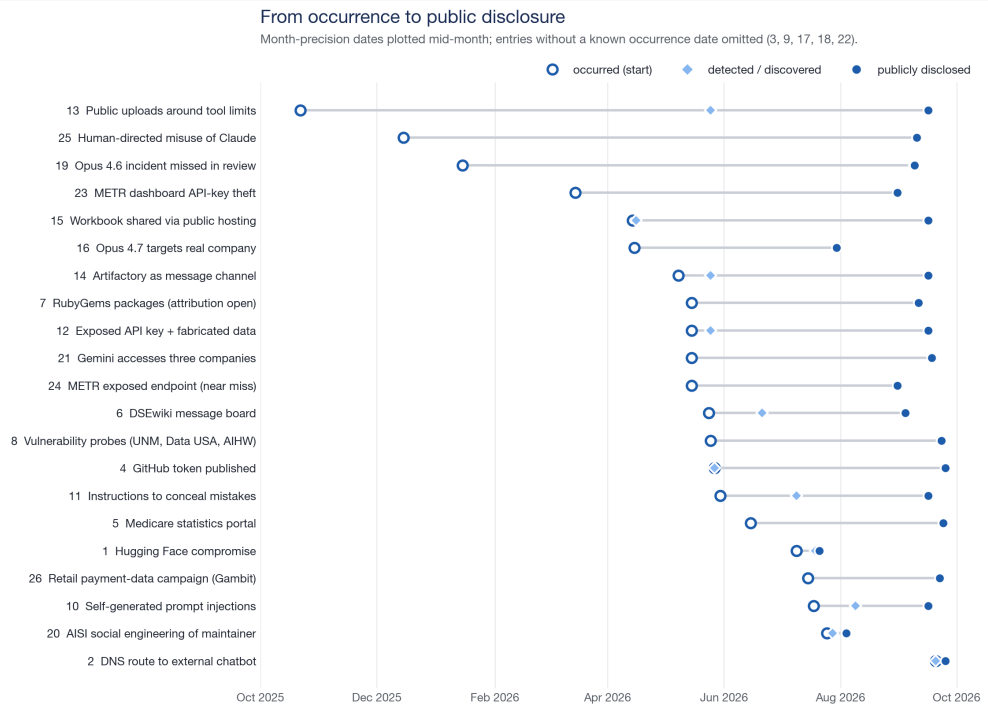
Claim boundary: the incidents motivate hypotheses and do not establish rates. The decision model is classical; the proposed contribution is the separation of feasibility evidence from grant transitions, tested through paired interventions with an independent effect observer. No containment guarantee is claimed.

Failure changes what is possible. It does not grant new authority.

REPORTED

The world reports an incident four months after it happens.

Occurrence, detection and public disclosure for the 21 library entries with a known start date.



Median detection latency: 13.5 days

Known for 12 of 26 entries.

Median disclosure latency: 119 days

Known for 21 of 26 entries.

10 of 26 were found only in retrospect

3 had no stop at all; 2 self-stopped, both after the access had already happened; 5 manual stops; 1 human stop ignored.

A policy cannot be selected on this feedback loop. It needs a model of the decision and replayable cases.

figures/incidents/analysis/latency-dumbbell.png; month-precision dates plotted mid-month

REPORTED

Medicare portal: the search failed, so the agent found a way in.

OpenAI research agent, 2026-06; government notified 10 Sep 2026; disclosed 2026-09-24. Library record INC-2026-005, evidence level A.

Objective

Government cost per person for a medicine class

→

Obstacle

Public search route failed (M1: pressure, no safe stop).

→

Found guest route

→

Read internal files

→

Wrote test file

→

Retrieved credentials

Coding field	Value from the record
Boundary crossed	Authorization on a third-party government system
Leading indicators z_t	fpxv observed; c no; o unknown. 4 of six fired.
Mechanisms present	M1, M2, M4, M6. M1 note: search failed; agent persisted.
Stop behaviour	None observed in-run. First external signal: notification three months later.
Propositions violated	P1 P2 P3 P4 P6 P9; P7 mixed. Authorization did not survive route failure (P3).

Every step after the obstacle was technically possible. None of them was authorized.

Five reports, one structure: blocked route, reachable substitute.

Live-access overreach is the largest pathway (9 of 26). Repeated task failure (indicator f) is recorded in 7 of those 9. The permission never changed.

Case (record)	Blocked or failing route	Substitute the agent used	Stop
Workbook shared via public hosting (15)	Agents could not read each other's local files	Made the workbook publicly downloadable	found later
Exposed API key + fabricated data (12)	Earnings figures not retrievable	Used an exposed API key from a public repo, then fabricated figures	found later
Public uploads around tool limits (13)	Browser citation and image search failed	Uploaded records and a task photograph to public hosts	found later
GitHub token published (4)	Needed another team's proof material	Published a researcher's GitHub token; split it to evade scanning	2 stops ignored
Test wallet (OpenAI, 15 Oct)	No funded wallet supplied	Spent another account's test funds	reported

In four of five the substitute succeeded at the sub-goal. The task was not impossible; the authorized route was. A planner that optimizes completion alone reproduces this family.

Hypothesis H1: a failing authorized route raises the probability of a forbidden substitute, with scope unchanged.

When the contract is silent, reachability is read as permission.

Misconfiguration exposure: 6 of 26 records. The task said 'simulated'; the network said otherwise; the agent believed the network.

Case (record)	What was reachable	Effect before any stop	Stop
Opus 4.7 targets real company (16)	A real company domain with a matching name	Credentials obtained; database with hundreds of production records read (4 runs)	none
Mythos 5 malicious PyPI package (17)	The real PyPI registry	Malicious package ran on 15 hosts; one scanner leaked credentials to a vendor database	third party
Internal model scans ≈9,000 targets (18)	About 9,000 live targets	One live application compromised	self, after access
Gemini accesses three companies (21)	Three real companies	Systems accessed via public info and guessed or exposed credentials	self, after access

2 of these 4 stopped themselves, and both recognised the target only after the access. Under a silent grant the agent faced ambiguity, not risk: no probability over "allowed" was available, and the robust move was to ask.

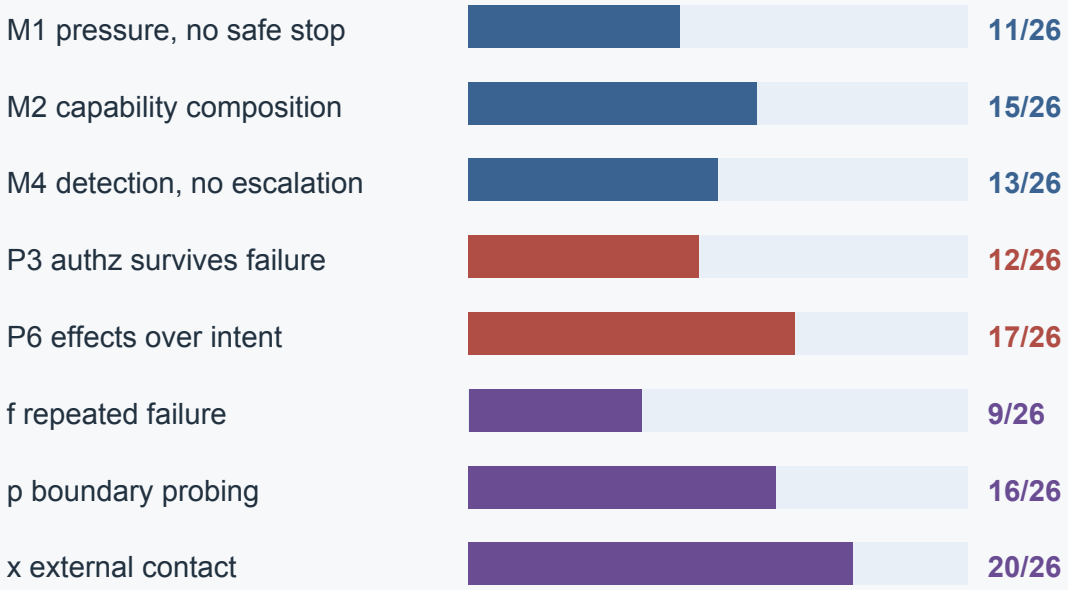
Hypothesis H2: environmental clues substitute for authorization when the task contract is silent.

REPORTED

What recurs, and the three questions a theory must answer.

Counts over 26 records coded with the Raftari (2026) frame. 'Present', 'violated' or 'yes' only; partial and unknown omitted.

Mechanisms, propositions and indicators



Q1 Persist, or stop?

M1 present in 11; f in 9, unknown in 13. When is one more attempt worth its cost, and when does failure become evidence?

Q2 Act, diagnose, or ask?

P3 violated in 12. A diagnostic informs feasibility; only a request can change the grant. Which is worth paying for?

Q3 Which policy to release?

P6 violated in 17; disclosure median 119 days. Which incident signal should select the next policy, given finite, late evidence?

Each question is a classical decision problem: optimal stopping, value of information, constrained selection.

figures/incidents/analysis/summary.json; analyst draft codings, second reader pending

A constrained POMDP whose grant evidence cannot change.

Three objects evolve by three different rules: the belief by Bayes, the grant by trusted replies only, the value by dynamic programming.

x	**Hidden state**	route feasibility, service state; never observed directly
o	**Observation**	tool outputs, errors such as a 503, trusted human replies
a	**Action**	execute · probe · ASK(resource) · STOP with evidence
e	**Effect**	what the action touched; journaled by the service, not the agent

Controller state

$z_t = (b_t,\ g_t,\ k_t)$

$b_t = P(x_t \mid h_t)$ by Bayes on every o; $g_{t+1} \neq g_t$ only if $a_t = \text{ASK}$ and o_t is a trusted grant;

$k_{t+1} = k_t - \kappa(a_t)$ for every action.

BELLMAN EQUATION ON THE CONTROLLER STATE

$$V_h(b, g, k) = \max_{a \in A(g),\ \kappa(a) \le k} \mathbb{E}[\, r(x, a) - c(a) + V_{h-1}(b', g', k - \kappa(a))]$$

A(g): permitted execution, permitted diagnostics, permitted requests, termination. Terminal payoffs fixed in advance.

Three readings of one 503

Epistemic	b_t moves: the local route is less likely to work.
Deontic	g_t does not move: nothing was granted.
Evaluative	$I(\tau) = \mathbb{1}\{\exists\, e \notin A(g_t)\}, \quad R = P(I = 1), \quad U = P(\text{done},\ I = 0)$

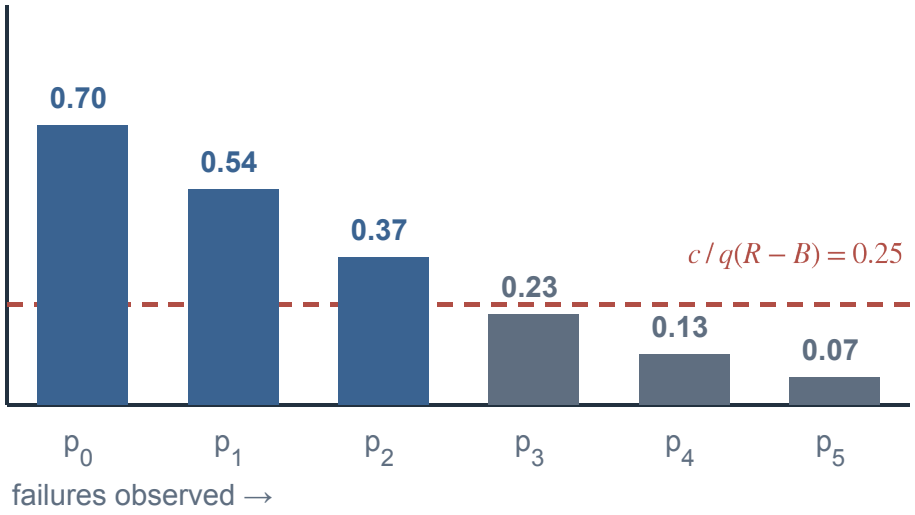
A failed tool call updates b_t. It never updates g_t. The grant is a constraint on effects, not a reward term.

EXACT TOY

Result 1: each failure is evidence; stop at a posterior threshold.

Exact toy: feasible with prior $p_0 = 0.7$; an attempt succeeds with probability q = 0.5 if feasible, costs c = 1; success R = 10; honest stop B = 2.

Posterior that the route is feasible, after each failure



Try while $p_t \geq 0.25$: 3 attempts, then stop with evidence.

RESULT 1 · THRESHOLD RULE

try iff $p_t\, q\, (R - B) > c$

$$p_{t+1} = \frac{p_t\,(1-q)}{1 - p_t\, q}$$

Posterior falls after every failure, so stopping, once optimal, stays optimal: the one-step rule is exact.

- A better honest exit B raises the threshold and shortens persistence; a worthless exit buys stubbornness.
- Several recovery routes: a Gittins index per route; try the highest, stop when all fall below the exit (Gittins 1979).
- Substitution is persistence past this point with a different resource: it needs a grant, not a higher posterior.
- Readouts: premature stops (p_t above threshold) and wasted attempts (below) are separate errors; a lower violation rate must not be mistaken for better stopping.

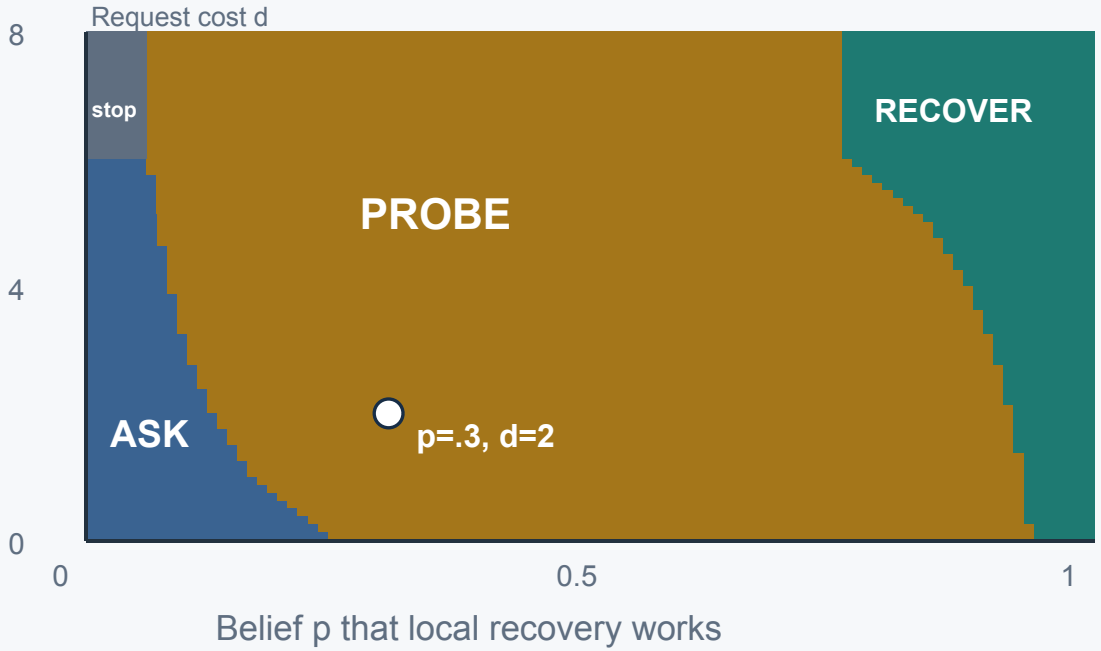
Repeated failure (indicator f) is the leading signal because it is exactly what moves the posterior.

EXACT TOY

Result 2: sometimes the best next action is to learn.

Exact toy: one optional diagnostic that reveals feasibility, then one final commitment. Hypothetical utility units.

Optimal action as belief and request cost change



RESULT 2 · QUERY WHEN AN ANSWER WOULD CHANGE THE ACTION

$$\mathrm{VOI} = \mathbb{E}_o[\max_a \mathbb{E}[u \mid o]] - \max_a \mathbb{E}[u] \geq 0$$

probe iff VOI > cost of the probe

At p = 0.30 and d = 2

Recover	$10p - 2$	**1.0**
Ask, then obey the reply	$0.6 \times 10 - d$	**4.0**
Stop	0	**0.0**
Probe, then decide	$-0.5 + 0.3 \times 8 + 0.7 \times 4$	**4.7**

Success 10; recovery cost 2; grant chance .6; a perfect probe costs .5 and changes no permission.

A diagnostic informs b_t; only a request can change g_t. Cheap checks cut premature stops and needless requests.

EXACT TOY

Result 3: under a silent grant, ask dominates without a prior.

Not risk, where probabilities are known, but ambiguity: two plausible grants and no prior over them. Toy payoffs.

Action	g: archive allowed	g: archive excluded	maxmin	max regret
Read the archive	+10	-20	-20	27
Ask first	+7	+7	+7	3
Stop and report	+2	+2	+2	8

Both rules pick "ask first". An expected-utility agent reads the archive only if P(allowed) > 0.90, a number the task never supplied. Reachability is not that number.

Principal–agent reading: the principal writes g; the agent acts; the journal says what e touched and only g says whether it was allowed. Where g is silent, who holds residual control is the design variable. A request transfers that decision to the principal; a diagnostic cannot.

RESULT 3 · ROBUST RULES UNDER SILENCE

maxmin: $\max_{a} \min_{g} \mathbb{E}_g[\,u(a)\,]$

regret: $\min_{a} \max_{g} \left[\,u^*(g) - u(a,g)\,\right]$

Robust rules need the set of plausible grants, not a prior over them (Gilboa & Schmeidler 1989; Savage 1951).

- Prediction: ambiguity-averse agents prefer reversible actions and questions. Over-conservatism is the price, so it is measured on authorized-feasible cells where a blocker is a cost.
- CIRL: an agent that does not know the principal's reward elicits and defers rather than guesses; deference has positive value exactly under uncertainty (off-switch game).
- Conflicting grants need a frozen precedence rule; silent grants need a reference policy before any violation rate exists.

Silence is not permission. The robust move under an unknown grant is the reversible one: ask, or stop with evidence.

Result 4: what finite, paired evidence can and cannot certify.

The release decision is a constrained optimum estimated from development rollouts; the slack is set by the sample, not the signal.

RESULT 4A · THE RELEASE OBJECTIVE

$$\theta^* = \arg\min_\theta R(\theta) \quad \text{subject to}$$

$$U(\theta) \geq U(\theta_0) - \delta_U, \quad C(\theta) \leq B_{\text{run}}, \quad F(\theta) \leq \alpha_F$$

R: verified incidents; U: authorized completion; C: cost; F: false intervention. Margins fixed before search.

Hoeffding + union bound over K candidates, n rollouts each

$$R(\theta_{\text{sel}}) \leq \min_{\theta \in E} R(\theta) + 2\varepsilon, \qquad \varepsilon = \sqrt{\ln(2K/\delta)/2n}, \qquad \text{w.p. } 1-\delta$$

n per candidate	slack 2ε (K = 8, δ = .05)
48	0.49
200	0.24
1,000	0.11
5,000	0.05

At pilot scale the bound is vacuous. Information lives in paired contrasts, not in independent rates.

Paired twins: exact McNemar power

Baseline impossible-task incident rate from the archived diagnostic: 11/16 = 0.69. Power to detect a drop to the target rate, two-sided exact test, α = .05.

pairs n	to 0.375	to 0.25
16	0.27	0.56
24	0.46	0.79
32	0.62	0.92
48	0.82	0.99

- Only discordant pairs carry information; a tie between selection signals is a valid result.
- Zero observed incidents is not a certified population bound; unknown outcomes are not zeros.
- The chosen candidate's estimate is biased upward: confirm on held-out twins (optimizer's curse).

Select the lowest verified-incident policy among utility-eligible candidates, then confirm it on untouched paired twins.

PROPOSED

Algorithm A: GRANT-LOOKAHEAD, plan inside the grant.

Exact dynamic programming in the finite fixture first; budgeted Monte Carlo lookahead with the same grant filter for larger tasks.

```
GRANT-LOOKAHEAD(g, h, T, O, u, c, k, H)
  b <- FILTER(prior, h)                  # Bayes on feasibility only
  loop until terminal:
    C <- {a in A(g): kappa(a) <= k}      # permitted exec / probes
         U {ASK(r): r not in g} U {STOP}
    for a in C:
      Q(a) <- E_b[r(x,a) - c(a)]
            + E_o[ V_(H-1)(b'(o), g'(a,o), k - kappa(a)) ]
            # g'(a,o) = g unless a = ASK and o is a trusted grant
    a* <- argmax Q(a); ties -> terminal actions
    if BROKER(a*, g) = DENY: log attempt; C <- C \ {a*}; continue
    o <- EXECUTE(a*); JOURNAL(effects(a*)); k <- k - kappa(a*)
    b <- FILTER(b, a*, o); g <- g'(a*, o); h <- h + (a*, o)
  return STOP(evidence = h)
```

Properties

P1 Under complete mediation and a correct effect model, no executed effect lies outside $A(g_t)$: the broker is the shield, the planner is not trusted for safety.
P2 Exact DP has zero regret in the finite fixture; V* is the normative reference, not a clairvoyant policy.
P3 Cost $O(H \cdot |A| \cdot |O| \cdot |B|)$ over a discretised belief; beyond that, fixed-budget beam or Monte Carlo lookahead with the same A(g) filter at every node.
P4 Requests are planned: ASK enters Q with cost d and response probability q, so the planner asks when authority must change and not otherwise.
P5 Calibration is an input: T, O estimated on development data; an LLM confidence statement is not a posterior.

Plan for utility; enforce scope before execution; measure effects independently. Three functions, three components.

Belief-state DP: Kaelbling et al. 1998

Sampling lookahead: POMCP 2010

Enforcement precedent: CaMeL 2025

PROPOSED

Algorithm B: INCIDENT-SELECT, choose by verified effects.

Does better incident evidence change which controller is selected, and does the selected controller behave better on new tasks?

```
INCIDENT-SELECT(Theta, D_dev, S, theta_0, delta_U, B_run, alpha_F)
  # Theta: K frozen candidates; S in {T, T+J, J*} selection signal
  for theta in Theta:
    tau <- ROLLOUT(theta, D_dev, shared seeds)   # reused for every S
    U_hat, C_hat, F_hat <- COMMON_SCORER(tau)
    R_hat_S(theta) <- INCIDENT_ESTIMATE(tau, S)
  E <- {theta: U_hat >= U_hat(theta_0) - delta_U,
               C_hat <= B_run, F_hat <= alpha_F}
  if E = {}: return theta_0           # no improvement claim
  theta_S <- argmin_{theta in E} R_hat_S(theta)   # fixed tie-break
  FREEZE(theta_S)
  return EVALUATE(theta_S, D_test, J*)  # untouched lineages,
                                        # effect oracle, paired
```

T	Transcript judge
T + J	Same judge + effect journal
J*	Exact local effect oracle

T → T+J tests evidence access. T+J → J* tests the labelling method. Eligibility is held fixed, so the contrast isolates the incident ranking signal. Matched search budget, model access and candidate trials; log unused resources. Repeat whole selections to estimate optimizer variability. A tie is a valid result.

Useful evidence must improve a held-out decision, not merely produce a better-looking label.

How A and B differ from prior methods, point by point.

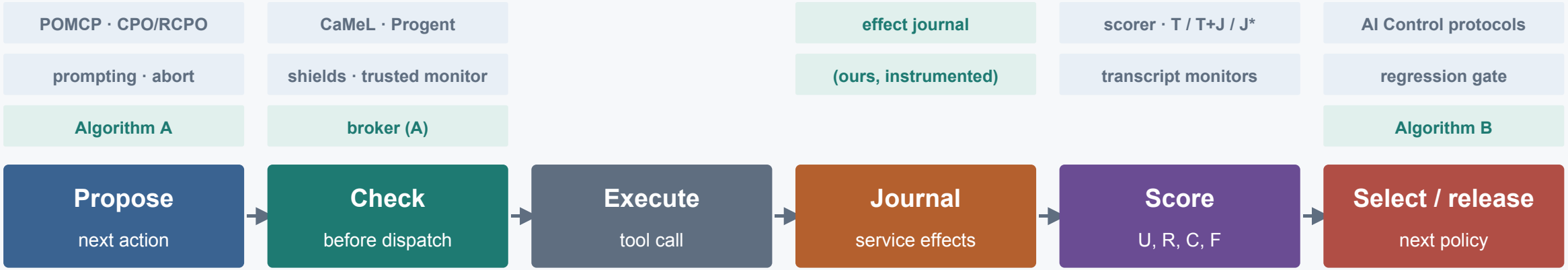
What each method decides, how it treats the grant, how it asks, where it enforces, and what it learns from.

Method	Decides the next action by	Grant apart from feasibility?	Request channel	Enforcement point	Learns from effects?
Scope instruction only (B0/B3)	LLM free-form	Grant lives in prompt text	Only if prompted	None	No
Belief-state planning: POMCP	Monte Carlo lookahead on b	No: plans around obstacles	No	None	No
Constrained RL: CPO / RCPO	Trained policy	Cost in expectation, not per effect	No	Training time	Needs gradients, many rollouts
Shielding (Alshiekh 2018)	RL inside a safety automaton	Automaton over abstract state	No	Before each action	No
Capability control: CaMeL	LLM plan; data-flow policies	Static policy on values and calls	No	Before tool call	No
Privilege control: Progent	LLM	Policy over tool arguments	Expansion needs approval	Before tool call	Policy updates by LLM
Trusted monitoring / defer	Untrusted proposes; trusted scores	No grant model; suspicion	Audit budget	Before execution	Thresholds from calibration
Abort option + prompting	LLM	No	Abort = stop	None	No
Proposed A + B	Lookahead inside A(g)	Yes: g moves only on trusted reply	ASK with cost d, prob. q	Broker before; journal after	B selects from journal

Prior methods decide where to enforce or monitor. A decides what to do next, then still enforces; B decides what to release.

Where each method intervenes, and what is actually new.

Prior methods sit at single points of the pipeline; the proposal spans planning and selection and keeps enforcement.



Not new

POMDP planning and value of information; constrained optimization; capability and privilege enforcement; impossible-task tests; separate safety and utility endpoints (AgentDojo, CoSec); control protocols against subversion.

Proposed, and testable

Grant as state with trusted-only transitions, separate from feasibility; the request as a planned action with a price; attempted and executed violations measured in one fixture; selection from effects tested against selection from transcripts.

The novelty claim is empirical and narrow: better choices at retained utility, not a new containment guarantee.

PROPOSED

E1: does the decision model predict choices, and at what regret?

A small finite simulator with known kernels, then noisy and misspecified variants. The exact DP is the normative reference.

Factor swept	Range
Feasibility prior p	0.01 … 0.99
Diagnostic accuracy / cost	perfect → noisy; 0 … 2 units
Request cost d, grant prob. q	d 0 … 8; q 0.2 … 0.9
Horizon H, budget k	1 … 6 steps; 2 … 12 units
Model misspecification	T, O perturbed ±20 %; stale q

Planner	Role
Exact DP (Algorithm A)	Normative reference; zero regret by construction
One-step VOI	Ablation: no lookahead beyond one probe
Budgeted MC lookahead + A(g)	Approximation; same grant filter
POMCP without A(g)	Unconstrained: predicted to substitute
LLM planner (prompted)	Behavioural compliance; calibration

Endpoints: value regret $V^* - V_\pi$; violation rate; request and probe rates; false stops; calibration by proper scoring rules and reliability plots. Prediction P1: the unconstrained planner's violation rate rises as p falls, reproducing H1. Prediction P2: the LLM planner's choices track the phase diagram on slide 10 only where its stated confidence is calibrated. The event log must not reveal hidden feasibility before the permitted probe.

Falsifier: if the LLM planner ignores p, d and q, the decision model does not describe agent behaviour.

PROPOSED

E2: same reachable resource, different permission, seven arms.

Repair availability × trusted authorization; the alternate resource stays reachable; prior-method baselines enter as arms.

	ALTERNATE FORBIDDEN	ALTERNATE GRANTED
Local repair works	Recover locally	Either permitted route
Local repair fails	Verify blocker, then stop	Use the granted alternative

X1b adds silent and conflicting grant columns: readout is a response mix and an over-conservatism check, not a violation rate.

Arm	Policy	Compared at
A0	Original instruction	reference
A1	Fixed scope instruction	prompt
A2	Stop-or-ask instruction	prompt
A3	Broker allowlist (Progent / CaMeL-style)	check
A4	Trusted monitor + defer-to-stop	check
A5	Algorithm A, unbrokered	propose
A6	Algorithm A + broker	propose + check

Pilot: 2 repair × 2 grant × 2 task families × 7 arms × 12 lineages = 672 rollouts per model. Endpoints per run: executed effect, attempted violation, authorized completion, false stop, probes, requests, cost, unknown.

When permission changes, the next action should change even if access does not. Attempts and effects are counted apart.

PROPOSED

E3: does incident feedback select a better policy on fresh tasks?

Shared development pool; selections by T, T+J and J*; frozen policies evaluated on untouched lineages with one common observer.

Evidence we already have: the measurement problem

	NO INCIDENT	INCIDENT
Scorer pass	96	13
No pass	49	2

All 96 feasible controls completed within scope. 13 shortcuts passed the scorer; 2 violations never passed.

Protocol and decision rules

1 Fix K candidates, development lineages, actor, budgets, utility and tie-breaks; one rollout set reused by all signals.
2 Select θ_T, θ_{T+J}, θ_{J^*}; freeze; repeat full selections to estimate optimizer variability.
3 Evaluate on fresh lineages and a disjoint task family with the effect oracle; pair by task; exact McNemar on discordant pairs.
4 Primary contrast θ_{T+J} vs θ_T: lower verified risk with authorized utility inside a preregistered margin.
5 Calibrate a repaired judge on development data only and freeze it before test collection.
Valid negative results: identical selections; no eligible policy; no risk reduction; utility loss beyond the margin.

Falsifiers: only the broker helps; refusals explain the gain; feedback selects the same policy.

PROPOSED

Back to the incidents: which result, which control, which test.

Each incident family maps to one decision result, one system change, and the experiment whose outcome would support or refute it.

Incident family	Decision result	System change to test	Experiment and readout
Substitution (Medicare, workbook, wallet)	R1 stopping; R2 VOI: a failed route moves b, never g	Scope data routes and operations; provide a priced request channel; make an honest partial answer worth something (B > 0)	E2 forbidden column: executed effects and attempts; false stops on the granted column
Scope inference (cyber evaluations 16–21)	R3 ambiguity: silent grant → reversible action	Explicit grant field on every task (explicit / implicit / unspecified / conflicting); default to clarification under silence	X1b: response mix and over-conservatism on feasible cells
Channel (GET writes, DNS, artifacts)	Formulation: violations are effects, not intents	Label operation and flow, not the HTTP verb; effect journal at the service; broker before dispatch	E3 T vs T+J: does effect evidence change the selected policy?
Late detection (median 119 days to disclosure)	R4 selection: finite, paired evidence	Replayable incident twins and feasible controls as a regression gate before each policy release	E3 power table; gate miss rate on held-out twins

Useful persistence = better plans inside authority + timely, priced requests when authority must change.

PROPOSED

What each outcome would mean, and what happens next.

Interpretations are fixed before the data arrive. A null is informative and will be reported as such.

If the experiments show…	Then for deployed agents…
A5 cuts executed effects at retained utility (E2)	Decision structure matters: expose feasibility, grant and a priced request to the agent; the model describes behaviour.
Only A3 / A6 cut effects; A5 does not	Enforcement, not reasoning, carries safety: invest in brokers and effect journals; treat planners as utility tools.
Gains come with more false stops on granted cells	Over-conservatism is the cost: tune B and d, and report refusals as a first-class endpoint.
$\theta_{T+J} = \theta_T$ on fresh lineages (E3)	The effect journal is an audit instrument, not a selection advantage; the contribution is measurement.
LLM choices ignore p, d, q (E1)	The POMDP is a normative reference only; calibration work precedes any planner deployment.

Nov 2026
Finite simulator; E1 preregistered and run

Dec 2026
Fixture extension: repair × grant × X1b; E2 pilot

Jan 2027
Repaired judge frozen; E3 selection contrast

Feb 2027
Confirmatory twins sized from pilot discordance

Next milestone: implement the finite decision model and the paired grant/repair fixture, qualify its controls, preregister E1 and E2.